



Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

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THE PROBLEM

Finance Today Is Broken

Modern payment systems are built on fragile assumptions — always-on internet, trusted third parties, and centralized data stores. Every transaction leaks metadata. Every POS terminal charges a premium. Every breach exposes everything.

Privacy Collapse

Every purchase is logged, tracked, and monetized. Centralized databases are single points of catastrophic failure.

Cloud Dependency

Offline environments render most fintech systems completely inoperable. Rural and low-connectivity markets are excluded.

POS Cost Burden

Monthly hardware and subscription fees make digital billing inaccessible for small merchants and micro-enterprises.

Data Breach Risk

A single server compromise exposes millions of users. Centralized architecture is structurally indefensible at scale.

THE SOLUTION

Introducing Spendly

Three interconnected applications — one unified ecosystem. Spendly reimagines peer-to-peer finance as a privacy-preserving, offline-capable, zero-infrastructure system built for the real world.

Spendly

The consumer wallet. Offline expense tracking, encrypted invoices, biometric authentication, and peer-to-peer settlement — all without a server.

Spendly Shop

The merchant POS. Barcode scanning, inventory management, QR invoicing, and cash wallet — zero monthly fees, zero internet required.

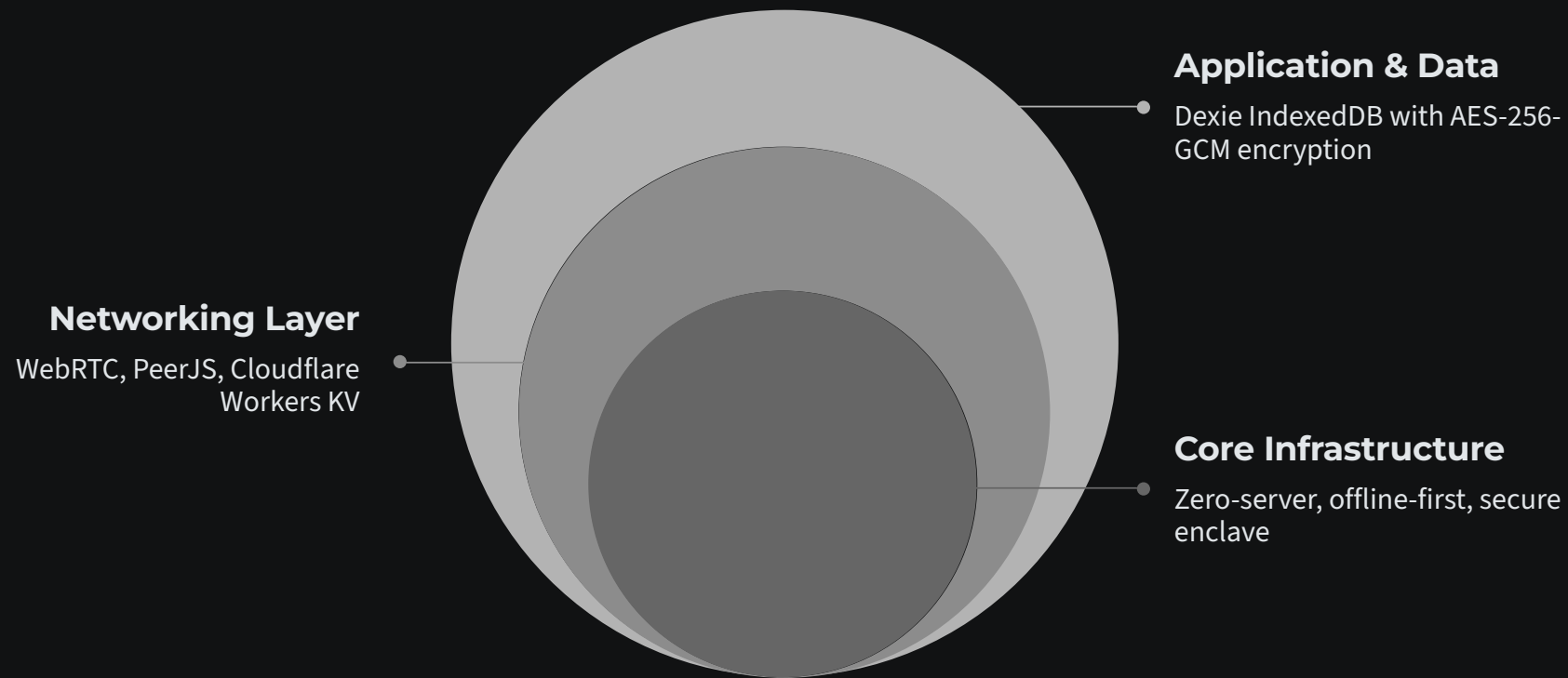
Spendly Radar

The discovery layer. Proximity-based offer detection, coupon claiming, and encrypted peer communication through WebRTC mesh networking.



System Architecture

Spendly's architecture is an exploded layered stack — each technology occupies a precise stratum, from the reactive interface down to cryptographic storage. No layer depends on a central authority.



React and PWA deliver the interface. IndexedDB with Dexie.js provides structured local persistence. WebRTC and PeerJS enable direct device-to-device communication. Cloudflare Workers with KV offer optional edge caching. WebAuthn and AES-256-GCM secure every layer — all without a central database.

WORKING FLOW

From Discovery to Settlement

01

Offer Discovery

Spendly Radar detects proximity-based merchant offers through encrypted peer signaling.

02

Coupon Claim & Barcode Scan

User claims offer; merchant scans product barcode via Spendly Shop for instant lookup.

03

Offer Validation & Invoice Generation

System validates eligibility, applies discount, and generates a cryptographically signed invoice.

04

Invoice Sharing & Expense Analytics

Invoice transfers peer-to-peer via WebRTC; expense is logged locally with real-time savings analytics.



The Technology Stack

Every technology in Spendly was selected for a single reason: it must work without a server. The stack is lean, modern, and purpose-built for offline-first cryptography.

Frontend & UI

- **React** — Component-driven reactive interface
- **PWA** — Installable, offline-capable web app
- **WebAuthn** — Biometric and hardware-key authentication

Database & Storage

- **IndexedDB** — Browser-native structured local storage
- **Dexie.js** — Ergonomic IndexedDB wrapper with schema versioning

Networking & Edge

- **WebRTC** — Direct peer-to-peer data channels
- **PeerJS** — Simplified WebRTC signaling abstraction
- **Cloudflare Workers** — Optional serverless edge functions
- **Worker KV** — Distributed key-value edge storage

Security

- **AES-256-GCM** — Authenticated encryption for all stored data
- **Secure Enclave** — Hardware-backed key isolation

SECURITY

Zero-Knowledge by Design

Spendly never transmits raw financial data. Every record is encrypted before storage. Every authentication is local. Every transaction is peer-to-peer. The server knows nothing — because there is no server.

Zero-Knowledge Architecture

No plaintext data ever leaves the device. The system proves validity without revealing content — mathematically, not trust-based.

Offline-First Encryption

AES-256-GCM encrypts all IndexedDB records at rest. Keys are derived from biometric credentials and never transmitted.

WebAuthn Biometric Unlock

FIDO2-compliant authentication binds access to hardware-backed biometrics. No passwords. No recovery emails. No attack surface.

Secure Enclave Isolation

Cryptographic keys are stored in hardware-isolated secure enclaves — inaccessible to the application layer or any remote adversary.

FEATURES

Everything a Modern Finance System Needs



Expense Tracking

Real-time local categorization with encrypted ledger history — fully operational without connectivity.



Merchant Billing

Instant QR invoice generation, barcode product lookup, and digital receipt delivery — zero hardware cost.



Inventory & Cash Wallet

Local stock management with cash balance tracking and EMI alert notifications for merchants.



Offline Sync & Analytics

Data syncs peer-to-peer when connectivity resumes. Spending insights are computed locally — never in the cloud.



What Spendly Delivers — and Where It Goes

Current Advantages

- **Offline-First** — Fully functional without internet
- **Zero Server Cost** — No infrastructure, no subscription
- **Privacy-Preserving** — Zero-knowledge, local-only data
- **Fast Billing** — Sub-second QR and barcode processing
- **P2P Communication** — Direct device-to-device settlement

Future Scope

- **AI Insights** — On-device spending pattern analysis
- **Voice Billing** — Voice-command invoice generation
- **UPI & NFC** — Native Indian payment rail integration
- **GST Compliance** — Automated tax invoice formatting
- **Multi-Store** — Chain merchant management layer



CONCLUSION

Spendly Is Not a Feature. It Is a Foundation.

Spendly demonstrates that privacy, offline capability, and peer-to-peer finance are not compromises — they are design imperatives. Built without a server, without a database, without a cloud. Built for the world as it actually exists.

Offline-First

Works everywhere, always

Zero-Knowledge

Your data stays yours

Peer-to-Peer

No middleman, ever

Thank you. Questions are welcome.